Kestrel Materials Testing Laboratory

Unit 4, Fenwick Trading Estate, Halden HD2 6PL
Accredited testing laboratory No. 4471

Test Report: Compressive Strength of Concrete Cubes

Client and project	
Client	Halden Civil Works Ltd
Project	Northgate Footbridge, abutment A
Client order	PO-55210

Specimens	
Specimen type	150 mm cubes, site-made
Curing	Water tank at 20 ± 2 °C
Test method	Procedure TP-07, issue 5

Report	
Report number	KMT-26-03187
Date of issue	21 October 2026
Specimens received	16 and 24 September 2026

Sample ref.	Pour date	Location	Age (days)	Test date	Specimen		Result			Failure mode
					Mass (kg)	Density (kg/m³)	Max. load (kN)	Strength (MPa)	Mean (MPa)	
C-301	15/09/2026	Abutment A footing, east bay	7	22/09/2026	8.12	2406	742.5	33.0	33.0	Satisfactory
C-302					8.09	2397	731.3	32.5		Satisfactory
C-303					8.15	2415	751.5	33.4		Satisfactory
C-304			28	13/10/2026	8.11	2403	1008.0	44.8	44.8	Satisfactory
C-305					8.14	2412	994.5	44.2		Satisfactory
C-306					8.07	2391	1023.8	45.5		Satisfactory
C-307	23/09/2026	Abutment A wall, lift 1	7	30/09/2026	8.05	2385	652.5	29.0	29.0	Satisfactory
C-308					8.10	2400	639.0	28.4		Unsatisfactory
C-309					8.02	2376	668.3	29.7		Satisfactory
C-310			28	21/10/2026	8.08	2394	895.5	39.8	39.7	Satisfactory
C-311					8.11	2403	870.8	38.7		Satisfactory
C-312					8.03	2379	913.5	40.6		Satisfactory

All specimens were received in good condition and were tested within the permitted tolerance on age. Strength is the maximum load divided by the nominal cross-sectional area of 22,500 mm².

The specified strength class is C35/45, so the 28-day results for both pours fall below the characteristic cube strength of 45 MPa.[1] Specimen C-308 showed an unsatisfactory failure pattern and its result should be treated with caution.

The results relate only to the specimens tested.

Tested by
R. Achterberg, Technician

Approved by
M. Sowande, Laboratory Manager

[1] Compliance is assessed by the client against clause 6.3 of the project specification.